SPLITTING THE DEFINING POLYNOMIALS OF A FINITE ALGEBRA AFTER A FINITE FREE BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- Each generator of a finite algebra satisfies a monic polynomial over the base. Algebra, Lemma \ref{algebra-lemma-finite-is-integral}, cited at proof lines 4597-4598. Provides the polynomials that are to be split.
- Adjoining a root of a monic polynomial inside a finite free extension. Lemma \ref{lemma-adjoin-one-root}, cited at proof lines 4600-4603. Produces the finite free extension in which all the polynomials split.

Source reference. Original lemma 0531: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6589–6589. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 00GK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 610–610. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0530: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6588–6588. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let $R \to S$ be a finite ring map. There exists a finite free ring extension $R \subset R'$ such that $S \otimes_R R'$ is a quotient of a ring of the form*

$$R'[T_1, \ldots, T_n]/(P_1(T_1), \ldots, P_n(T_n))$$

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with $P_i(T) = \prod_{j=1,\ldots,d_i}(T - \alpha_{ij})$ *for some* $\alpha_{ij} \in R'$.

Proof. Let $x_1, \ldots, x_n \in S$ be generators of S over R. For each i we can choose a monic polynomial $P_i(T) \in R[T]$ such that $P_i(x_i) = 0$ in S, see Algebra, Lemma 00GK. Say $\deg(P_i) = d_i$. By Lemma 0530 (applied $\sum d_i$ times) there exists a finite free ring extension $R \subset R'$ such that each P_i splits completely:

$$P_i(T) = \prod_{j=1,\ldots,d_i}(T - \alpha_{ij})$$

for certain $\alpha_{ik} \in R'$. Let $R'[T_1, \ldots, T_n] \to S \otimes_R R'$ be the R'-algebra map which maps T_i to $x_i \otimes 1$. As this maps $P_i(T_i)$ to zero, this induces the desired surjection. □